

Coupled Impulsive Control for Multisynchronization of Multistable Stochastic Neural Networks Under Parameter Uncertainties

Xuemei Li[✉], Song Zhu[✉], Zhen Zhang[✉], Weiwei Luo[✉], and Shiping Wen[✉], *Senior Member, IEEE*

Abstract—Multisynchronization of multistable stochastic neural networks (MSNNs) is investigated herein. First, an MSNN multistable stochastic neural network (SNN) model subject to both time-varying delays and parameter uncertainties is constructed, which accurately describes neural dynamics in realistic environments. Subsequently, to reduce control costs, a coupled impulsive control strategy is adopted for studying the multisynchronization problem among MSNN systems. By developing an appropriate Lyapunov functional and leveraging the average impulsive interval concept, sufficient conditions for multisynchronization of the considered SNNs under both fixed and switching topologies are derived. Finally, the effectiveness of the proposed control scheme is verified through a numerical example.

Index Terms—Coupled impulsive control, multistable stochastic neural networks (MSNNs), multisynchronization, parameter uncertainties, time-varying delays.

I. INTRODUCTION

ARTIFICIAL neural networks (NNs), owing to their powerful nonlinear modeling capabilities and adaptive learning mechanisms, have become fundamental tools in addressing core problems such as complex system optimization, pattern recognition, and intelligent decision-making [1], [2], [3]. Their unparalleled advantages in processing high-dimensional, nonlinear, and uncertain data have been further demonstrated in recent years [4], [5]. Among diverse neural architectures, stochastic neural networks (SNNs) have attracted growing research interest due to their inherent ability to model the intrinsic and extrinsic randomness prevalent in real-world systems [6], [7]. Consequently, analyzing the dynamic behavior of SNNs and ensuring their robustness has become an important theoretical basis for the reliability of intelligent systems.

Received 10 September 2025; revised 10 December 2025; accepted 26 February 2026. Date of publication 10 March 2026; date of current version 2 September 2026. This work was supported in part by the Key Program of the National Natural Science Foundation of China under Grant 62333016; and in part by the National Natural Science Foundation of China under Grant 62173177, Grant 62276119, and Grant 62373358. (Corresponding author: Song Zhu.)

Xuemei Li, Song Zhu, Zhen Zhang, and Weiwei Luo are with the School of Mathematics, JCAM, China University of Mining and Technology, Xuzhou 221116, China (e-mail: ts24080015a31@cumt.edu.cn; songzhu@cumt.edu.cn; zhenzhang013@126.com; weiweiluo@cumt.edu.cn).

Shiping Wen is with the Faculty of Engineering and Information Technology, Australian Artificial Intelligence Institute, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: shiping.wen@uts.edu.au).

Digital Object Identifier 10.1109/TNNLS.2026.3669910

In practice, however, the dynamics of neural networks are inevitably subject to multiple disturbances [8], [9], including time-varying delays [10], [11] and parameter uncertainties [12]. These factors pose significant challenges to maintaining stability and consistent performance, potentially causing network states to deviate from expected trajectories and even inducing undesirable phenomena such as chaotic oscillations or synchronization failures. Therefore, it is imperative to establish rigorous dynamical analysis frameworks.

In the dynamic behavior of the studied neural network, the multisynchronization problem of NNs has attracted increasing attention from scholars owing to the wide range of applications such as engineering control, secure communication, neuroscience, and other fields [13]. Synchronization [14], [15], as a fundamental phenomenon in natural and engineering systems, enables coordinated behavior between distributed units, such as neuron spike synchronization in biological networks and clock alignment in communication protocols. In contrast to traditional synchronization, where the entire network converges to a single state [16], [17], [18], [19], multisynchronization describes a collective dynamical behavior in which multistable subnetworks simultaneously synchronize onto multiple distinct target trajectories or manifolds. This phenomenon, which typically arises in large-scale systems like convolutional neural networks, enables complex information processing and parallel computing [13], particularly enhancing multimodal signal fusion and multitask coordination. Therefore, achieving and controlling multisynchronization is considered a core mechanism enabling efficient parallel computing and bionic intelligence.

Up to date, a variety of control methodologies have been developed to achieve synchronization in neural networks, including adaptive control [20], feedback control [19], event-triggered control [21], [22], [23], and impulsive control [24], [25], [26], [27]. Among them, impulsive control has been widely applied due to its efficiency, low energy consumption, and ease of implementation [28], [29], [30]. By applying instantaneous state resets at specific discrete time points, impulsive control can effectively regulate network dynamics while reducing control costs. Notably, Wang et al. [31] pioneered the study of both dynamic multisynchronization (DMS) and static multisynchronization (SMS) in impulsively controlled coupled neural networks. Building upon this, Lv et al. [32] established a unified impulsive control framework capable of analyzing both DMS and SMS. Further extending

the applicability, [33] incorporated time delays and introduced an event-triggered impulsive control strategy to achieve exponential synchronization in delayed networks. Most recently, Liu et al. [34] extended these results by applying event-triggered impulsive control to address multisynchronization in coupled networks.

Despite significant advances, a critical discrepancy persists between theoretical developments and the demands of real-world applications. Recent research has made considerable progress in various aspects of synchronization control, while these challenges are often addressed in isolation. For instance, most studies on multisynchronization in coupled multistable networks (see [31], [32]) are confined to deterministic frameworks, thus limiting their applicability in stochastic environments. Furthermore, while event-triggered impulsive control schemes such as [34] achieve multisynchronization, they overlook the ubiquitous presence of time-varying delays inherent in practical neural and engineering systems. Concurrently, robust impulsive and synchronization methods have been developed to handle uncertainties and delays (see [12], [14], [16], [24]). However, these are primarily designed for mono-synchronization, where the entire network converges to a single state, and thus fail to exploit the parallel processing capabilities offered by multistable systems. Additional limitations persist in several existing frameworks, including [31], which rely on restrictive assumptions such as the differentiability of time-varying delays that are often impractical to satisfy or verify in real-world implementations. Moreover, many control schemes are validated only under fixed network topologies, despite the fact that switching topologies are more representative of dynamic modern networks, such as mobile sensor arrays or reconfigured computing clusters.

To overcome these limitations, we construct a multistable SNN (MSNN) model with parameter uncertainties. Under specified assumptions and interval division, the network possesses 2^n mean-square locally exponentially equilibrium state. Then, impulsive control is employed to achieve multisynchronization, with the primary objective of analyzing the existence of multiple synchronous manifolds in stochastic environments. The principal contributions of this work are highlighted below.

- 1) A novel hybrid uncertain neural network model is developed, integrating time-varying delays, stochastic disturbances, and parameter uncertainties. In contrast to prior models that consider only deterministic influences or a single stochastic factor [32], [33], [34], the proposed model more faithfully captures the complex dynamical environment of real neural systems, thereby substantially improving the generality and practical applicability of neural network modeling.
- 2) Multisynchronization is realized in MSNNs, enabling different subnetworks to converge to distinct synchronous manifolds in a coordinated manner. Unlike traditional single synchronization schemes [12], [17], [25], the proposed approach supports parallel multitask processing and significantly enhances the efficiency of complex computational tasks.
- 3) By combining Lyapunov stability theory with the concept of average impulsive interval, sufficient con-

ditions are derived for achieving multisynchronization under stochastic perturbations and uncertain parameters. Another advantage of the obtained results lies in the removal of restrictive differentiability assumptions on time-varying delays, with applicability extended to switching network topologies. In comparison with existing approaches such as [31], this provides a more flexible and theoretically sound framework for analyzing complex multisynchronization phenomena in SNNs.

The structure of the remaining part of this article is outlined below. Section II establishes notations and preliminary concepts. Building on this foundation, Section III develops impulsive control methods for both dynamical and SMS in MSNNs. Verification through numerical simulations is presented in Section IV. Finally, Section V concludes this article with main findings and future research directions.

II. PRELIMINARY

This section introduces the necessary preliminaries from graph theory [35] and presents the model for SNNs.

Let $\mathbb{R}$ be the set of real numbers, and $\mathbb{R}_+$ denote the non-negative reals. Let $\mathcal{V} = (\mathbb{R}^n, \|\cdot\|)$ be the normed vector space of real n -vectors equipped with the Euclidean metric. The space of linear transformations from $\mathbb{R}^m$ to $\mathbb{R}^n$ is isomorphic to $\mathbb{R}^{n \times m}$. Let $\mathbb{Z}_+$ represent the positive integers, and $\Lambda = \{1, 2, \dots, n\}$. For a square matrix $\mathcal{M}$, let $\lambda_{\max}(\mathcal{M})$ and $\lambda_{\min}(\mathcal{M})$ denote the largest and smallest eigenvalues of $\mathcal{M}$, respectively. The notation $\mathcal{M} > 0$ indicates that $\mathcal{M}$ is symmetric and positive definite (respectively, $\mathcal{M} < 0$ is negative definite). $\mathcal{E} \otimes \mathcal{F}$ denotes the Kronecker product of matrices $\mathcal{E}$ and $\mathcal{F}$. The $n \times n$ identity matrix is written as E_n , and $\mathbf{1}_N$ is the N -dimensional vector of ones. Let $\mathcal{M}$ and $\mathcal{N}$ be subsets of $\mathbb{R}^k$ (with $1 \leq k \leq n$). We introduce the following notations:

$$\begin{aligned} C(\mathcal{M}, \mathcal{N}) &= \{\varphi : \mathcal{M} \rightarrow \mathcal{N} \mid \varphi \text{ is a continuous mapping}\} \\ C^1(\mathcal{M}, \mathcal{N}) &= \{\varphi : \mathcal{M} \rightarrow \mathcal{N} \mid \varphi \text{ is continuously differentiable}\}. \end{aligned}$$

In the context of symmetric matrices, the symbol $\star$ is used to denote symmetric blocks.

Before building the model, we provide the background of graph theory.

A. Graph Theory [35]

The communication topology is modeled through $\mathcal{P} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where $\mathcal{P}$ is a directed graph and vertex collection $\mathcal{V} = \{1, 2, \dots, N\}$ contains all nodes, arc set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ defines interconnection links, and weight matrix $\mathcal{A} = [a_{ij}]_{N \times N}$ quantifies connection strengths with nonnegative entries $a_{ij} \geq 0$. An interconnection from vertex v_j to v_i exists precisely when (v_j, v_i) belongs to $\mathcal{E}$, corresponding to $a_{ij} > 0$. Self-connections are explicitly prohibited through the condition $a_{ii} = 0$ for all vertices. Each computational entity corresponds to a node, with direct data transfer occurring along established edges. A directed sequence of connected vertices forms an oriented pathway within $\mathcal{P}$. A subconfiguration $\mathcal{T}_{\mathcal{P}} = (\mathcal{V}_{\mathcal{T}}, \mathcal{E}_{\mathcal{T}})$ constitutes a spanning arborescence when it satisfies three conditions: $\mathcal{V}_{\mathcal{T}} = \mathcal{V}$, directed tree structure, and existence of a root vertex with zero in-degree that connects to all other

vertices through unique directed chains. The vertex degree matrix $\mathbf{D} = \text{diag}\{d_1, \dots, d_N\}$ is determined by the expression $d_i = \sum_{j \in \mathcal{N}(i)} a_{ij}$, while the graph Laplacian operator $\mathbf{L}$ is defined through $\mathbf{L} = \mathbf{D} - \mathcal{A}$.

Additionally, the Laplacian operator $\mathbf{L}$ possesses the null row-sum property, making $\mathbf{1}_N$ its eigenvector for eigenvalue zero. This eigenvalue becomes algebraically simple, and all other eigenvalues exhibit positive real components if and only if the directed structure contains a spanning arborescence.

B. Model Description

This work examines a group of interconnected neural networks modeled by following stochastic differential equations:

$$\begin{aligned} dx_i(t) = & \left[-(\mathcal{C} + \Delta\mathcal{C})x_i(t) + (\mathcal{A} + \Delta\mathcal{A})g(x_i(t)) \right. \\ & \left. + (\mathcal{B} + \Delta\mathcal{B})g(x_i(t - \tau(t))) + I(t) \right] dt \\ & + \phi_i(t, x(t), x(t - \tau(t))) d\omega(t). \end{aligned} \quad (1)$$

The state of subnetwork i at time t is described by the vector $x_i(t) \in \mathbb{R}^n$ for $i = 1, 2, \dots, N$, where $N \geq 2$ denotes the number of identical subnetworks in the system. $\mathcal{C} = \text{diag}\{c_1, \dots, c_n\} > 0$ is the self-feedback matrix, with positive diagonal elements $c_j > 0$ representing each neuron's self-regulation strength. Connection matrices $\mathcal{A} = [a_{lj}]_{n \times n}$ and $\mathcal{B} = [b_{lj}]_{n \times n}$ govern inter-neuronal communication, where matrix elements a_{lj} and b_{lj} quantify instantaneous and delayed interaction strength, respectively, between neuron pairs (l, j) within subnetwork i , with all neuronal indices contained in $\Lambda = \{1, \dots, n\}$. Neuronal activation functions $g_j : \mathbb{R} \rightarrow \mathbb{R}$ collectively form the vector $g(x_i(t)) = [g_1(x_{i1}(t)), \dots, g_n(x_{in}(t))]^\top$. Parametric uncertainties are modeled through unknown matrices $\Delta\mathcal{C}, \Delta\mathcal{A}, \Delta\mathcal{B} \in \mathbb{R}^{n \times n}$. The system includes a continuous periodic function $I : \mathbb{R} \rightarrow \mathbb{R}^n$ and features bounded transmission delay $\tau(t)$ satisfying $0 \leq \tau(t) \leq \bar{\tau}$ for constant $\bar{\tau} > 0$. On the complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, stochastic effects arise from a 1-D standard Brownian motion $\omega(t)$ is defined, with neuronal diffusion properties governed by the coefficient $\phi : \mathbb{R}^+ \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Remark 1: In practical applications, neural networks are inevitably affected by parameter fluctuations and external interference. Therefore, in exploring the problem of multisynchronization, this study comprehensively considers random interference and parameter uncertainty, making the model more applicable and more in line with practical engineering scenarios. Specifically, compared with the research in [31], [32], [33], and [34] that mainly focus on single-factor analysis, the modeling framework and research scope of this work are more comprehensive in terms of factor coverage and more systematic in terms of analysis logic.

Remark 2: The theoretical pursuit of multisynchronization is strongly motivated by its profound potential in engineering systems operating within complex, stochastic environments. For instance, in cooperative multirobot search and rescue systems, distinct subteams must maintain their own synchronous formation patterns (DMSs) or converge to different task-specific locations (SMSs) despite stochastic communication delays and sensor noise. Similarly, in neuromorphic computing, different processor clusters may need to synchronize

to distinct computational rhythms while handling stochastic workload fluctuations.

In such scenarios, the ability to achieve and sustain multiple synchronous patterns, rather than merely a single global state, is essential. The inherent stochastic disturbances and time delays render existing deterministic or single-pattern synchronization schemes inadequate. Advancing the multisynchronization framework for stochastic networks, as undertaken in this article, therefore represents not merely a theoretical extension but a critical step toward upgrading neural networks from handling a single task to performing multiple tasks simultaneously in complex, uncertain environments, while demonstrating strong anti-interference capabilities and high energy efficiency.

C. Notations

To ensure the existence of equilibrium points, we present the following proposition.

Proposition 1 [36]: Given input $I \in \mathbb{R}^n$, any solution $\bar{x} \in \mathbb{R}^n$ to the equilibrium equation

$$-(\mathcal{C} + \Delta\mathcal{C})\bar{x} + [(\mathcal{A} + \Delta\mathcal{A}) + (\mathcal{B} + \Delta\mathcal{B})]g(\bar{x}) + I = 0$$

yields $\phi(t, \bar{x}, \bar{x}) = 0$.

By Proposition 1, for each input vector I , $x^* \in \mathbb{R}^n$ is an equilibrium of MSNNs if and only if

$$\begin{aligned} & -(\mathcal{C} + \Delta\mathcal{C})x^* + (\mathcal{A} + \Delta\mathcal{A})g(x^*) \\ & + (\mathcal{B} + \Delta\mathcal{B})g(x^*) + I = 0. \end{aligned}$$

Then, let us consider the issue of state space partition. $J(1) = (1, +\infty)$ denotes the positive half-line, $J(-1) = (-\infty, -1)$ represents the negative half-line, and $\{\pm 1\}^n$ the n -dimensional Cartesian product space of $\{-1, 1\}$. For arbitrary sign vectors $\zeta = (\zeta_1, \dots, \zeta_n)^\top \in \{\pm 1\}^n$ and inputs $I \in \mathbb{R}^n$, we define rectangular domains

$$\Delta_\zeta = \prod_{i=1}^n J(\zeta_i) = J(\zeta_1) \times \dots \times J(\zeta_n).$$

Remark 3: This work investigates MSNNs, which inherently require each subnetwork to possess multiple equilibrium states. As established in [36], the state-space partitions into $2^n + 1$ distinct regions, with 2^n unbounded rectangular subdomains. These rectangular regions exhibit positive invariance, and every single one of them includes a unique equilibrium point. Moreover, this equilibrium point is locally exponentially stable when evaluated in the mean-square sense.

Consider $R(t)$ as a solution trajectory within the set $\{R_l(t)\}_{l=1}^{2^n}$, where each $R_l(t)$ corresponds to either a locally exponentially stable equilibrium point or a stable periodic orbit in the time-varying delayed MSNNs described by system (1).

Definition 1 [31]: Dynamical multisynchronization (DMS) in MSNNs is achieved when the following criteria are satisfied below.

- 1) For arbitrary initial states $x(\zeta) = [x_1^\top(\zeta), \dots, x_N^\top(\zeta)]^\top$ with $x_i(\zeta) \in C([-\tau, 0], \mathbb{R}^n)$ for $i = 1, \dots, N$, there exists a synchronized trajectory $X_l^*(t) \in \mathbb{R}^n$ such that $\lim_{t \rightarrow \infty} x_i(t) = X_l^*(t) \quad \forall l \in \{1, 2, \dots, 2^n\}$, where l indexes

distinct synchronization patterns. The synchronization manifold associated with the initial state is characterized by the collective pattern $1_N \otimes X_l^*(t)$.

- 2) No fewer than two distinct initial configurations $x'(\zeta)$ and $x''(\zeta)$ are guaranteed to yield different synchronization manifolds $1_N \otimes X_l^*(t)$ and $1_N \otimes X_k^*(t)$ with the separation property $\inf_{t \geq t_0} \|1_N \otimes X_l^*(t) - 1_N \otimes X_k^*(t)\| > 0$. Additionally, for each manifold, there exists a δ -neighborhood $\mathcal{N}_\delta = \{x(t) \in \mathbb{R}^{Nn} : 0 < \|x(t) - 1_N \otimes X_l^*(t)\| < \delta\}$ containing trajectory points not belonging to the manifold itself.

Assumption 1: Assume the activation function $g_i(\cdot)$ satisfies a global Lipschitz condition, meaning there exist positive constants l_i fulfilling

$$|g_i(\zeta) - g_i(\xi)| \leq l_i |\zeta - \xi|$$

for all $\zeta, \xi \in \mathbb{R}$, where $|\cdot|$ denotes the absolute value and $i = 1, 2, \dots, n$. The diagonal matrix $L_f = \text{diag}\{l_1^g, l_2^g, \dots, l_n^g\}$ is subsequently defined based on these Lipschitz constants.

To characterize the temporal distribution of control impulses, the concept of average impulsive interval is introduced.

Definition 2 [37]: Consider an impulsive sequence $\theta = \{t_1, t_2, \dots\}$ satisfying $0 = t_0 < t_1 < t_2 < \dots$ with $\lim_{k \rightarrow \infty} t_k = \infty$. The average impulsive interval T_a for θ is defined by the existence of positive constants $N_0 \in \mathbb{Z}_+$ and $T_a > 0$ such that

$$N_\theta([t, T]) \geq \frac{T - t}{T_a} - N_0 \quad \forall T \geq t \geq 0.$$

Here, $N_\theta([t, T])$ denotes the count of impulsive events in the sequence θ occurring within the time interval $[t, T]$.

Then, consider the necessary assumptions for system uncertainty and random disturbances.

Assumption 2: Suppose k_0, k_1 , and k_2 are three given positive constants. For the matrices $\Delta\mathcal{C}$, $\Delta\mathcal{A}$, and $\Delta\mathcal{B}$ satisfy

$$\Delta\mathcal{C}^\top \Delta\mathcal{C} \leq k_0 E_n, \Delta\mathcal{A}^\top \Delta\mathcal{A} \leq k_1 E_n, \Delta\mathcal{B}^\top \Delta\mathcal{B} \leq k_2 E_n.$$

Assumption 3 [16]: The function $\phi(t, x(t), x_\tau(t))$ exhibits local Lipschitz continuity and linear growth behavior, satisfying $\phi(t, 0, 0) = 0$ at origin. Additionally, two matrices Θ_1 and Θ_2 satisfy

$$\begin{aligned} & \text{Tr}[\phi^\top(t, x_i(t), x_i(t - \tau(t))) \phi(t, x_i(t), x_i(t - \tau(t)))] \\ & \leq \|\Theta_1^\top x(t)\|^2 + \|\Theta_2^\top x(t - \tau(t))\|^2. \end{aligned}$$

Assumption 4 [31]: There exists a spanning tree in the directed graph $\mathcal{P}$ whose edges follow a unified direction.

Assumption 5 [31]: For every index $i \in \{1, 2, \dots, J\}$, the directed graph $\mathcal{P}_i$ possesses a spanning tree where all edges are directed away from the root node.

To facilitate the subsequent stability analysis, we recall several fundamental mathematical results that will be instrumental in deriving our main theorems.

Lemma 1 [38]: Consider symmetric matrices P and S with $P^\top = P$, $S^\top = S$, and an arbitrary matrix Q . The partitioned matrix inequality

$$\Pi = \begin{bmatrix} P & Q \\ Q^\top & S \end{bmatrix} < 0 \quad (2)$$

holds precisely when any one of the following equivalent conditions is fulfilled.

- 1) P is negative definite and $S - Q^\top P^{-1} Q$ is negative definite.
- 2) S is negative definite and $P - QS^{-1}Q^\top$ is negative definite.

Lemma 2 [39]: Consider the time-delay functional $G(t, u(t), u(t - \tau(t))) : \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ and impulsive operators $I_k : \mathbb{R} \rightarrow \mathbb{R}$. Assume G maintains nondecreasing behavior with respect to its third argument for fixed $(t, u(t))$, while each I_k is globally nondecreasing. For solutions $u(t)$ and $v(t)$ satisfying

$$\begin{cases} D_t^+ u(t) - G(t, u(t), u_\tau(t)) \leq 0, & t \neq t_k \\ u(t_k) - I_k(u(t_k^-)) \leq 0, & k \geq 1 \end{cases}$$

and

$$\begin{cases} D_t^+ v(t) - G(t, v(t), v_\tau(t)) > 0, & t \neq t_k \\ v(t_k) - I_k(v(t_k^-)) \geq 0, & k \in \mathbb{N} \end{cases}$$

with $D_t^+ u(t) = \overline{\lim}_{g \rightarrow 0^+} [u(t + g) - u(t)]/g$. If $u(t) \leq v(t)$ holds on $[-\bar{\tau}, 0)$, then this ordering persists for all $t > 0$.

III. MAIN RESULTS

A. MSNNs With Fixed Topology

Applying impulsive control to MSNNs (1) under fixed topology yields

$$\begin{aligned} dx_i(t) = & [-(\mathcal{C} + \Delta\mathcal{C})x_i(t) + (\mathcal{A} + \Delta\mathcal{A})g(x_i(t)) \\ & + (\mathcal{B} + \Delta\mathcal{B})g(x_i(t_\tau)) + I(t) + u_i(t)] dt \\ & + \phi_i(t, x(t), x_\tau(t)) d\omega(t) \end{aligned} \quad (3)$$

where $u_i(t) \in \mathbb{R}^n$ represents control input

$$u_i(t) = \hat{d} \left[\sum_{\substack{m=1, \\ m \neq i}}^N a_{im} (x_m(t) - x_i(t)) \right] \delta(t - t_k). \quad (4)$$

Within the interval $[t_k, t_{k+1}]$ for each $k \in \mathbb{R}_+$, the coupling gain $\hat{d} > 0$ must be designed at time t . For the directed graph $\mathcal{P}$, elements of its weighted adjacency matrix are expressed as a_{im} , while the Dirac delta function is denoted by $\delta(\cdot)$.

Furthermore, we can obtain the compact form of (3)

$$\begin{cases} dx(t) = [-(E_N \otimes (\mathcal{C} + \Delta\mathcal{C}))x(t) \\ \quad + (E_N \otimes (\mathcal{A} + \Delta\mathcal{A}))G(x(t)) \\ \quad + (E_N \otimes (\mathcal{B} + \Delta\mathcal{B}))G(x_\tau(t)) + (E_N \otimes E_n) \\ \quad \times I(t)] dt + \Phi(t, x(t), x_\tau(t)) d\varpi, \quad t \neq t_k \\ \Delta x(t_k) = -(\hat{d} \mathbf{L} \otimes E_n)x(t_k^-), \quad k \in \mathbb{Z}_+ \end{cases} \quad (5)$$

where $x(t) = [x_1^\top(t), x_2^\top(t), \dots, x_N^\top(t)]^\top$, $G(x(t)) = [g^\top(x_1(t)), g^\top(x_2(t)), \dots, g^\top(x_N(t))]^\top$, and $G(x_\tau(t)) = [g^\top(x_1(t - \tau(t))), \dots, g^\top(x_N(t - \tau(t)))]^\top$. The digraph $\mathcal{P}$ is characterized by its Laplacian matrix $\mathbf{L}$. Impulsive state transitions satisfy $\Delta x(t_k) = x(t_k) - x(t_k^-)$, and the left-hand limit denoted as $x(t_k^-) = \lim_{\epsilon \rightarrow 0^+} x(t_k - \epsilon)$. Additionally, the Kronecker product expression $\varpi = E_N \otimes \omega(t)$ represents stochastic effects.

Introducing the transformation $y_i(t) = x_i(t) - R(t)$, the controlled MSNNs with fixed topology and time-varying delays (5) undergo reformulation to yield

$$\begin{cases} dy(t) = \left[- (E_N \otimes (\mathcal{C} + \Delta\mathcal{C})) y(t) \right. \\ \quad + (E_N \otimes (\mathcal{A} + \Delta\mathcal{A})) G(y(t)) \\ \quad + (E_N \otimes (\mathcal{B} + \Delta\mathcal{B})) G(y_\tau(t))] dt \\ \quad + \Phi(t, y(t), y_\tau(t)) d\varpi, \quad t \neq t_k \\ \Delta y(t_k) = -(\hat{d}L \otimes E_n) y(t_k^-), \quad k \in \mathbb{Z}_+ \end{cases} \quad (6)$$

where $y(t) = [y_1^\top(t), y_1^\top(t), \dots, y_N^\top(t)]^\top$ represents the composite state vector. The mapping satisfies $G(y(t)) = G(x(t)) - G(R(t))$.

Furthermore, let us introduce the consensus error $e_i(t) = y_i(t) - y_N(t)$ for $i = 1, \dots, N-1$, and set $g(e_i(t)) = g(y_i(t)) - g(y_N(t))$, $g_\tau(y_N(t)) = g(y_N(t - \tau(t)))$, $\mathcal{C}_\infty = \mathcal{C} + \Delta\mathcal{C}$, $\mathcal{A}_\infty = \mathcal{A} + \Delta\mathcal{A}$, $\mathcal{B}_\infty = \mathcal{B} + \Delta\mathcal{B}$, $L_1 = [l_{ij} - l_{Nj}]_{(N-1) \times (N-1)}$, $i, j \in \{1, 2, \dots, N-1\}$, $L_2 = [l_{N1}, l_{N2}, \dots, l_{NN-1}]$.

Then the dynamics of the synchronization error (6) admit the following equivalent representation:

$$\begin{cases} \begin{bmatrix} de(t) \\ dy_N(t) \end{bmatrix} = \left(- \begin{bmatrix} E_N \otimes \mathcal{C}_\infty & 0 \\ 0 & \mathcal{C}_\infty \end{bmatrix} \begin{bmatrix} e(t) \\ y_N(t) \end{bmatrix} \right. \\ \quad + \begin{bmatrix} E_N \otimes \mathcal{A}_\infty & 0 \\ 0 & \mathcal{A}_\infty \end{bmatrix} \begin{bmatrix} G(e(t)) \\ g(y_N(t)) \end{bmatrix} \\ \quad + \begin{bmatrix} E_N \otimes \mathcal{B}_\infty & 0 \\ 0 & \mathcal{B}_\infty \end{bmatrix} \begin{bmatrix} G_\tau(e(t)) \\ g_\tau(y_N(t)) \end{bmatrix} \Big) dt \\ \quad + \begin{bmatrix} \Phi(t, e(t), e_\tau(t)) & 0 \\ 0 & \Phi(t, y_N(t), y_N(t_\tau)) \end{bmatrix} \\ \quad \times \begin{bmatrix} d\varpi \\ d\omega \end{bmatrix}, \quad t \neq t_k \\ \begin{bmatrix} \Delta e(t) \\ \Delta y_N(t) \end{bmatrix} = -\hat{d} \begin{bmatrix} L_1 & 0 \\ L_2 & 0 \end{bmatrix} \otimes E_n \begin{bmatrix} e(t_k^-) \\ y_N(t_k^-) \end{bmatrix}, \quad k \in \mathbb{Z}_+. \end{cases}$$

For each $i = 1, 2, \dots, N-1$, consider the error variable $e_i(t) \triangleq y_i(t) - y_N(t)$. It follows that $e_i(t) = [x_i(t) - R(t)] - [x_N(t) - R(t)] = x_i(t) - x_N(t)$. Set $\tilde{\omega} = E_{N-1} \otimes \omega(t)$, the synchronization behavior of system (6) directly reflects the stability properties of the transformed dynamical system described as follows:

$$\begin{cases} de(t) = \left[- (E_{N-1} \otimes (\mathcal{C} + \Delta\mathcal{C})) e(t) \right. \\ \quad + (E_{N-1} \otimes (\mathcal{A} + \Delta\mathcal{A})) G(e(t)) \\ \quad + (E_{N-1} \otimes (\mathcal{B} + \Delta\mathcal{B})) G(e_\tau(t))] dt \\ \quad + \Phi(t, e(t), e_\tau(t)) d\tilde{\omega}, \quad t \neq t_k \\ \Delta e(t_k) = -(\hat{d}L_1 \otimes E_n) e(t_k^-), \quad k \in \mathbb{Z}_+. \end{cases} \quad (7)$$

Then, let us consider the initial state of system (7) $e_i(t) = \xi_i(t)$, $t \in [-\tau, 0]$. Here, each $\xi_i \in C([-\tau, 0], \mathbb{R}^n)$ represents an initial error function that is characterized by the norm $\|\xi_i\|_\tau = \sup_{t \in [-\tau, 0]} |\xi_i(t)|$.

Under the premise that all the above conditions have been explicitly established, the following theorems of the delayed MSNNs are presented below.

Theorem 1: Consider the delayed multisynchronization problem of the MSNNs described in (3). Under Assumptions 1–4, and given an impulsive sequence $\zeta = t_1, t_2, \dots$, with an

average interval T_a , the system achieves DMS if there exist a matrix $H > 0$ and two diagonal matrices $Q_1, Q_2 > 0$, and positive scalars $\varepsilon_0, \varepsilon_1, \varepsilon_2$ such that the following inequalities hold for the designed control parameters:

$$\begin{bmatrix} -\alpha E_{N-1} & (E_{N-1} - \hat{d}L_1)^\top \\ \star & -E_{N-1} \end{bmatrix} \leq 0 \quad (8)$$

$$\begin{bmatrix} \Pi & H\mathcal{A} & H & H & H\mathcal{B} & H & \Theta_1 \\ \star & -Q_1 & 0 & 0 & 0 & 0 & 0 \\ \star & \star & -\varepsilon_0 E_n & 0 & 0 & 0 & 0 \\ \star & \star & \star & -\varepsilon_1 E_n & 0 & 0 & 0 \\ \star & \star & \star & \star & -Q_2 & 0 & 0 \\ \star & \star & \star & \star & \star & -\varepsilon_2 E_n & 0 \\ \star & \star & \star & \star & \star & \star & -\frac{1}{\rho} E_n \end{bmatrix} < 0 \quad (9)$$

$$\begin{bmatrix} \Sigma & \Theta_2 \\ \star & -\frac{1}{\rho} E_n \end{bmatrix} < 0, \quad (10)$$

$$\delta + \frac{\ln \alpha}{T_a} + \gamma \alpha^{-N_0} < 0 \quad (11)$$

where

$$\begin{aligned} \Pi &= -H\mathcal{C} - \mathcal{C}^\top H + L_f(k_1 \varepsilon_1 E_n + Q_1)L_f + k_0 \varepsilon_0 E_n - \delta H \\ \Sigma &= L_f(k_2 \varepsilon_2 E_n + Q_2)L_f - \gamma H. \end{aligned}$$

Proof: The proof establishes multisynchronization through three main steps. First, constructing a Lyapunov function to analyze stability between subsystems. Then, handling impulsive effects through matrix inequalities. Finally, applying the comparison principle to derive convergence conditions.

Take $V(t)$ as the Lyapunov function where

$$V(t) = e^\top(t) (E_{N-1} \otimes H) e(t). \quad (12)$$

Here, $V(t)$ is chosen to measure the synchronization error between $N-1$ subsystems, where $H > 0$ provides weighting for the error states.

Applying Itô's formula, the generator $\mathbf{L}V(t)$ accounts for both the deterministic drift and stochastic diffusion components, which are computed as

$$\begin{aligned} \mathbf{L}V(t) &= 2e^\top(t) (E_{N-1} \otimes H) \dot{e}(t) \\ &\quad + \frac{1}{2} \text{Tr} \left[\Phi^\top(t, e, e_\tau) V_{ee}(t, e) \Phi(t, e, e_\tau) \right] \\ &= -2e^\top(t) (E_{N-1} \otimes H) (\mathcal{C} + \Delta\mathcal{C}) e(t) \\ &\quad + 2e^\top(t) (E_{N-1} \otimes H) (\mathcal{A} + \Delta\mathcal{A}) G(e(t)) \\ &\quad + 2e^\top(t) (E_{N-1} \otimes H) (\mathcal{B} + \Delta\mathcal{B}) G(e_\tau) \\ &\quad + \text{Tr} \left[\Phi^\top(t, e, e_\tau) (E_{N-1} \otimes H) \Phi(t, e, e_\tau) \right]. \end{aligned} \quad (13)$$

To deal with the parameter uncertainties, we estimate the following terms using Young's inequality which introduces tuning parameters $\varepsilon_0, \varepsilon_1, \varepsilon_2 > 0$ to balance the estimation tightness. For the perturbation term involving $\Delta\mathcal{C}$, we have

$$\begin{aligned} &-2e^\top(t) (E_{N-1} \otimes H \Delta\mathcal{C}) e(t) \\ &\leq \varepsilon_0^{-1} e^\top(t) (E_{N-1} \otimes HH) e(t) \\ &\quad + \varepsilon_0 e^\top(t) (E_{N-1} \otimes \Delta\mathcal{C}^\top \Delta\mathcal{C}) e(t) \end{aligned}$$

$$\begin{aligned} &\leq \varepsilon_0^{-1} e^\top(t) (E_{N-1} \otimes HH) e(t) \\ &\quad + k_0 \varepsilon_0 e^\top(t) (E_{N-1} \otimes I_n) e(t) \end{aligned} \quad (14)$$

where $k_0 > 0$ is a known constant bounding $\Delta \mathcal{C}^\top \Delta \mathcal{C} \leq k_0 E_n$.

Similarly, by Young's inequality and Lipschitz property, we can obtain

$$\begin{aligned} &2e^\top(t) (E_{N-1} \otimes H\mathcal{A}) G(e(t)) \\ &\leq e^\top(t) (E_{N-1} \otimes H\mathcal{A}Q_1^{-1}\mathcal{A}^\top H) e(t) \\ &\quad + e^\top(t) (E_{N-1} \otimes L_f Q_1 L_f) e(t) \end{aligned} \quad (15)$$

$$\begin{aligned} &2e^\top(t) (E_{N-1} \otimes H\Delta\mathcal{A}) G(e(t)) \\ &\leq \varepsilon_1^{-1} e^\top(t) (E_{N-1} \otimes HH) e(t) \\ &\quad + k_1 \varepsilon_1 e^\top(t) (E_{N-1} \otimes L_f^2) e(t) \end{aligned} \quad (16)$$

$$\begin{aligned} &2e^\top(t) (E_{N-1} \otimes H\mathcal{B}) G(e_\tau(t)) \\ &\leq e^\top(t) (E_{N-1} \otimes H\mathcal{B}Q_2^{-1}\mathcal{B}^\top H) e(t) \\ &\quad + e^\top(t_\tau) (E_{N-1} \otimes L_f Q_2 L_f) e_\tau(t) \end{aligned} \quad (17)$$

$$\begin{aligned} &2e^\top(t) (E_{N-1} \otimes H\Delta\mathcal{B}) G(e_\tau(t)) \\ &\leq \varepsilon_2^{-1} e^\top(t) (E_{N-1} \otimes HH) e(t) \\ &\quad + k_2 \varepsilon_2 e^\top(t_\tau) (E_{N-1} \otimes L_f^2) e_\tau(t) \end{aligned} \quad (18)$$

where $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$ are given constants, and Q_1 and Q_2 are positive definite diagonal matrices. Using Assumption 2, the diffusion term satisfies

$$\begin{aligned} &\text{Tr}[\Phi^\top(t, e, e_\tau) (E_{N-1} \otimes H) \Phi(t, e, e_\tau)] \\ &\leq \rho [e^\top(t) \Theta_1 \Theta_1^\top e(t) + e^\top(t_\tau) \Theta_2 \Theta_2^\top e_\tau(t)]. \end{aligned} \quad (19)$$

which effectively couples the stochastic energy with deterministic bounds through the scalar ρ . Combining (14)–(19), we collect all upper bounds

$$\begin{aligned} \mathbf{L}V(t) &\leq e^\top(t) (E_{N-1} \otimes \Xi_1) e(t) \\ &\quad + e^\top(t_\tau) (E_{N-1} \otimes \Xi_2) e_\tau(t) \end{aligned} \quad (20)$$

where $\Xi_1 = -H\mathcal{C} - \mathcal{C}^\top H + H\mathcal{A}Q_1^{-1}\mathcal{A}^\top H + H(\varepsilon_0^{-1}E_n + \varepsilon_1^{-1}E_n + \varepsilon_2^{-1}E_n)H + H\mathcal{B}Q_2^{-1}\mathcal{B}^\top H + L_f(k_1\varepsilon_1 E_n + Q_1)L_f + k_0\varepsilon_0 E_n + \rho\Theta_1\Theta_1^\top$, $\rho = \lambda_{\max}(H)$ and $\Xi_2 = \rho\Theta_2\Theta_2^\top + L_f(k_2\varepsilon_2 E_n + Q_2)L_f$.

If inequalities (9) and (10) hold, it follows that:

$$\mathbf{L}V(t) \leq \delta V(t) + \gamma V(t_\tau) \quad (21)$$

indicating that between impulses, the Lyapunov function decays exponentially with rate δ , while the delayed term introduces a scaled perturbation $\gamma V(t_\tau)$.

Observation of (8) provides the matrix inequality

$$\begin{aligned} &\begin{bmatrix} -\alpha E_{N-1} & (E_{N-1} - \hat{d}L_1)^\top \\ \star & -E_{N-1} \end{bmatrix} \leq 0 \\ &\quad \updownarrow \\ &-\alpha E_{N-1} + (E_{N-1} - \hat{d}L_1)^\top (E_{N-1} - \hat{d}L_1) \leq 0 \end{aligned}$$

which leads to the jump inequality

$$\begin{aligned} V(t_k) &= e^\top(t_k) (E_{N-1} \otimes H) e(t_k) \\ &= e^\top(t_k^-) \left[(E_{N-1} - \hat{d}L_1)^\top (E_{N-1} - \hat{d}L_1) \otimes H \right] e(t_k^-) \\ &\leq \alpha V(t_k^-), \quad \alpha < 1. \end{aligned} \quad (22)$$

Given an arbitrary positive scalar $\varepsilon > 0$, the corresponding impulsive system with time delay possesses a unique solution

denoted by $v(t)$. The dynamics of the auxiliary scalar comparison system are described by

$$\begin{aligned} \dot{v}(t) &= \delta v(t) + \gamma v(t_\tau) + \varepsilon, \quad t \neq t_k \\ v(t_k) &= \alpha v(t_k^-), \quad k \in \mathbb{Z}^+ \\ v(p) &= \lambda_{\max}(H) \mathbb{E}[\|\phi(p)\|^2], \quad p \in [-\tau, 0]. \end{aligned}$$

Observing that $\mathbb{E}[V(t)] \leq v(t)$ holds for $t \in [-\tau, 0]$, Lemma 2 implies $0 \leq \mathbb{E}[V(t)] \leq v(t)$ when $t > 0$, which allow us to analyze the simpler system $v(t)$ instead of the original system. Applying the parameters variation method, the solution takes the form

$$v(t) = F(t, 0) v(0) + \int_0^t F(t, p) [\gamma v(p - \tau(p)) + \varepsilon] dp \quad (23)$$

where $F(t, p)$ serves as the Cauchy matrix for

$$\begin{cases} \dot{\gamma}(t) = \delta \gamma(t), & t \neq t_k \\ \gamma(t_k) = \alpha \gamma(t_k^-), & k \in \mathbb{N}. \end{cases} \quad (24)$$

Using the average impulsive interval T_a , the Cauchy matrix upper bound satisfies

$$\begin{aligned} F(t, p) &= e^{\eta(t-p)} \prod_{p \leq t_k \leq t} \alpha \\ &\leq \alpha^{\frac{t-p}{T_0} - N_0} \exp(\eta(t-p)) \\ &= \alpha^{-N_0} \exp[(\delta + \ln \alpha / T_a)(t-p)]. \end{aligned} \quad (25)$$

According to inequalities (23) and (25), and set $\mu = \alpha^{-N_0} \lambda_{\max}(H) \mathbb{E}[\|\phi_i(p)\|^2]$, one can derive the following upper bound:

$$\begin{aligned} v(t) &\leq \mu \exp \left[\left(\delta + \frac{\ln \alpha}{T_a} \right) t \right] \\ &\quad + \int_0^t \alpha^{-N_0} \exp \left[\left(\delta + \frac{\ln \alpha}{T_a} \right) (t-p) \right] \\ &\quad \times [\gamma v(p - \tau(t)) + \varepsilon] ds. \end{aligned} \quad (26)$$

Define the function $f(\lambda) = \lambda + \delta + \frac{\ln \alpha}{T_a} + \alpha^{-N_0} \gamma e^{\lambda \tau}$. Observe that $f(0) < 0$, while $f(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow +\infty$. Moreover, since the derivative $f'(\lambda) = 1 + \tau \alpha^{-N_0} \gamma e^{\lambda \tau}$ is always strictly positive, the function $f(\lambda)$ is strictly increasing. Hence, there exists a single positive solution $\rho > 0$ such that $f(\rho) = 0$ where ρ characterizes the actual synchronization convergence rate. That is,

$$f(\rho) = \rho + \delta + \frac{\ln \alpha}{T_a} + \alpha^{-N_0} \gamma e^{\rho \tau} = 0.$$

Thus, it is established that

$$\begin{aligned} v(t) &= \lambda_{\max}(H) E[\|\phi_i(p)\|^2] \\ &< \mu < \mu e^{-\rho t} + \frac{\varepsilon}{M}, \quad -\tau \leq t \leq 0 \end{aligned}$$

where $M = -(\delta + ((\ln \alpha)/T_a))\alpha^{-N_0} - \gamma$ satisfies $M > 0$. We now establish the inequality

$$v(t) < \mu e^{-\rho t} + \frac{\varepsilon}{M}, \quad t > 0. \quad (27)$$

Suppose, contrary to (27), there exists $t' > 0$ satisfying

$$v(t') \geq \mu e^{-\rho t'} + \frac{\varepsilon}{M} \quad (28)$$

while for all $t \in (0, t')$

$$v(t) \leq \mu e^{-\rho t} + \frac{\varepsilon}{M}.$$

Define the parameters

$$\sigma = -\left(\delta + \frac{\ln \alpha}{T_\alpha}\right), \quad \chi = \gamma \alpha^{-N_0}.$$

The solution satisfies the following bound:

$$\begin{aligned} v(t') &\leq \mu e^{-\sigma t'} + \int_0^{t'} \alpha^{-N_0} e^{-\sigma(t'-p)} [\gamma v(p - \tau(t)) + \varepsilon] dp \\ &\leq \mu e^{-\sigma t'} + \int_0^{t'} \frac{e^{-\sigma(t'-p)}}{\alpha^{N_0}} \left[\gamma \left(\mu e^{-\rho(p-\tau)} + \frac{\varepsilon}{M} \right) \right] dp \\ &\quad + \varepsilon \int_0^{t'} \frac{e^{-\sigma(t'-p)}}{\alpha^{N_0}} dp \\ &\leq \mu e^{-\sigma t'} + \alpha^{-N_0} e^{-\sigma t'} \int_0^{t'} e^{-(\rho-\sigma)p} \gamma \mu e^{\rho \tau} dp \\ &\quad + \left(\frac{\varepsilon \gamma}{M} + \varepsilon \right) \int_0^{t'} \alpha^{-N_0} e^{-\sigma(t'-p)} dp \\ &\leq \mu e^{-\sigma t'} + \mu e^{-\sigma t'} \int_0^{t'} e^{-(\rho-\sigma)p} [-(\rho-\sigma)] dp \\ &\quad + \frac{\varepsilon(\gamma + M)}{M} \int_0^{t'} \alpha^{-N_0} e^{-\sigma(t'-p)} dp \\ &= \mu e^{-\sigma t'} + \mu e^{-\rho t'} - \mu e^{-\sigma t'} + \frac{\varepsilon}{M} (1 - e^{-\sigma t'}) \\ &= \mu e^{-\rho t'} + \frac{\varepsilon}{M} (1 - e^{-\sigma t'}) \\ &\leq \mu e^{-\rho t'} + \frac{\varepsilon}{M}. \end{aligned}$$

This contradicts assumption (28), confirming that inequality (27) must hold. Taking the limit as $\varepsilon \rightarrow 0$, then

$$\mu e^{-\lambda t} \geq v(t) \geq \mathbb{E}[V(t)] \quad \forall t \in \mathbb{R}_{\geq 0}.$$

Consequently, the asymptotic behavior satisfies

$$\mathbb{E}(\|e(t)\|^2) \leq \mu e^{-\lambda t} \sup_{p \in [-\tau, 0]} \mathbb{E}(\|\phi(p)\|^2) \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

This implies state convergence

$$\lim_{t \rightarrow \infty} x_i(t) = x_j(t) \quad \forall i, j \in \{1, \dots, N\}.$$

Thus, system (3) achieves complete synchronization for arbitrary initial conditions.

Additionally, examine the dynamics of the N th subnetwork

$$\begin{cases} dx_N(t) = [-\mathcal{C} + \Delta \mathcal{C}] x_N(t) + (\mathcal{A} + \Delta \mathcal{A}) g(x_N(t)) \\ \quad + (\mathcal{B} + \Delta \mathcal{B}) g(x_N(t_\tau)) + I(t) \\ \quad + \phi(t, x(t), x_\tau(t)) d\omega(t), \quad t \neq t_k \\ \Delta x_N(t_k) = \sum_{j=1}^{N-1} a_{Nj} (x_j(t_k^-) - x_N(t_k^-)), \quad k \in \mathbb{N}. \end{cases} \quad (29)$$

From (29), we deduce that $\Delta x_N(t_k) \rightarrow 0$ asymptotically. Consequently, the first equation in (29) characterizes the N th subnetwork's dynamical evolution. Our analysis reveals that this subnetwork possesses 2^n distinct periodic solutions, each distinct and locally exponentially stable, represented by $\mathcal{R} = \{R_k(t)\}_{k=1}^{2^n}$. Asymptotically, all states converge to a common trajectory $\lim_{t \rightarrow \infty} x_i(t) = R_l(t)$, $i \in \{1, 2, \dots, N\}$,

$l \in \{1, 2, \dots, 2^n\}$. Thus, DMS of MSNNs (3) is achieved under fixed topology. This completes the proof. $\square$

Remark 4: Compared with the existing literature, the results reported in this article form a substantial generalization and a unified theoretical framework. First, by removing the stochastic perturbations, our criteria degenerate exactly to the deterministic multistability conditions in [32], which shows that the proposed method retains the full strength of the classical results while operating in a more general stochastic setting. Second, when parameter uncertainties are further eliminated, the obtained conclusions naturally include those of [31] as special cases. More importantly, the new framework jointly considers stochastic disturbances and parametric uncertainties, which have not been simultaneously addressed in [31] and [32]. Therefore, the proposed results not only unify but also extend the existing works in a strictly more general and practically relevant direction.

Remark 5: It is necessary to reveal the different roles and practical significance of control parameters α , δ , and γ in achieving multisynchronization of MSNNs. These parameters collectively determine the convergence speed, impulse strength, average trigger interval, and robustness of the system. The impulse control strength coefficient $\alpha < 1$ quantifies the state error compression at impulsive instants, where smaller values of α enhance convergence speed but may increase sensitivity to noise and induce oscillations. The state decay rate δ governs the exponential decay of current state errors during continuous dynamics; larger δ requires stronger stability margins; thus, the system becomes more sensitive to model uncertainties and disturbances. The delay influence coefficient γ reflects the impact of delayed states on system dynamics, where larger γ values slow convergence and may cause instability under significant time-varying delays. These parameters are mathematically coupled through the sufficient condition (11), revealing a quantitative tradeoff: reducing α compensates for larger δ or γ , while in delay-dominated systems, priority should be given to reducing γ or enhancing impulsive control via smaller α .

Assuming a constant input vector I of appropriate dimensions where $I(t) \equiv I$, SMS for the controlled delayed MSNNs (4) can be established in accordance with Theorem 1, as formalized below.

Corollary 1: Consider the delayed MSNNs (5) under a constant input $I(t) \equiv I$. Suppose Assumptions 1–4 are satisfied, and the impulsive sequence has an average interval T_a . The system achieves SMS if there exist a matrix $H > 0$, diagonal matrices $Q_1, Q_2 > 0$, and positive scalars $\varepsilon_0, \varepsilon_1, \varepsilon_2$ such that conditions (8)–(11) hold.

B. MSNNs With Switching Topology

Considering MSNNs that have switching topologies, the control law is given by

$$u_i(t) = \hat{d}^{p(t)} \left[\sum_{\substack{m=1 \\ m \neq i}}^N a_{im}^{p(t)} (x_m(t) - x_i(t)) \right] \delta(t - t_j)$$

where $t \in [t_k, t_{k+1})$ for $k \in \mathbb{N}$. Here, $p(t) : [0, +\infty) \rightarrow \{1, 2, \dots, l\}$ represents a switching signal that is

right-continuous; $\hat{d}^{p(t)} > 0$ denotes the configurable coupling strength; and $a_{im}^{p(t)}$ correlates with elements in the weighted adjacency matrix of digraph $\mathcal{P}_{p(t)}$.

Consequently, the MSNNs with impulsive control exhibit the following dynamics:

$$\begin{cases} dx(t) = [-(E_N \otimes (\mathcal{C} + \Delta\mathcal{C}))x(t) \\ \quad + (E_N \otimes (\mathcal{A} + \Delta\mathcal{A}))G(x(t)) \\ \quad + (E_N \otimes (\mathcal{B} + \Delta\mathcal{B}))G(x_\tau(t)) \\ \quad + (E_N \otimes E_n)E(t)]dt \\ \quad + \Phi(t, x(t), x_\tau(t))d\varpi, \quad t \neq t_k \\ \Delta x(t_k) = -(\hat{d}^{p(t)} \mathbf{L}^{p(t)} \otimes E_n)x(t_k^-), \quad k \in \mathbb{Z}_+ \end{cases} \quad (30)$$

Theorem 2: Under Assumptions 1–3 and 5, the MSNNs with switching topologies achieve DMS if there exist a matrix $H > 0$, diagonal matrices $Q_1, Q_2 > 0$, positive scalars $\varepsilon_0, \varepsilon_1, \varepsilon_2$, and control parameters including the topology-dependent coupling gains $\hat{d}^{p(t)} > 0$ adapting to switching Laplacians $L_1^{p(t)}$, the impulsive suppression factor $\alpha < 1$ governing instantaneous error reduction, the Lyapunov decay rate δ , and the delay robustness coefficient γ , such that for an impulsive sequence ζ with average interval T_a , the matrix inequalities (9)–(11) and the switching topology condition satisfy

$$\begin{bmatrix} -\alpha E_{N-1} & (E_{N-1} - \hat{d}^{p(t)} L_1^{p(t)})^\top \\ \star & -E_{N-1} \end{bmatrix} \leq 0. \quad (31)$$

Proof: Define $V(t) = e^\top(t)(E_{N-1} \otimes H)e(t)$. Then, from (31), we obtain the matrix inequality

$$-\alpha E_{N-1} + (E_{N-1} - \hat{d}^{p(t)} L_1^{p(t)})^\top (E_{N-1} - \hat{d}^{p(t)} L_1^{p(t)}) \leq 0.$$

The distinctive feature of switching topology analysis concerns the treatment of impulsive instants where topology transitions coincide with control impulses. At each switching instant t_k , the communication topology transitions from $\mathcal{P}_{p(t_k)}$ to $\mathcal{P}_{p(t_k^-)}$, while the state evolution obeys

$$\begin{aligned} V(t_k) &= e^\top(t_k^-) \left[(I_{N-1} - \hat{d}^{p(t_k)} L_1^{p(t_k)})^\top \right. \\ &\quad \times (I_{N-1} - \hat{d}^{p(t_k)} L_1^{p(t_k)}) \otimes H \left. \right] e(t_k^-) \\ &\leq \alpha V(t_k^-) \end{aligned} \quad (32)$$

with $\alpha < 1$. By employing the analytical framework established in Theorem 1, it can be deduced that the delayed MSNNs attain DMS under the proposed switching control. $\square$

Considering a constant input vector I of appropriate dimensions where $I(t) \equiv I$, we establish SMS for the controlled delayed MSNNs (30) based on Theorem 2, as formalized below.

Corollary 2: For the delayed MSNNs (30) under constant input I , assume the existence of a matrix $H > 0$ and diagonal matrices $Q_1, Q_2 > 0$. Suppose Assumptions 1–3 and 5 are satisfied and have average interval T_a . If there exist positive scalars $\gamma, \delta, \varepsilon_0, \varepsilon_1, \varepsilon_2$, a scalar $\alpha < 1$, and $\hat{d}^{p(t)} > 0$ ($p(t) = \{1, 2, \dots, l\}$) such that conditions (9)–(11), and (31), then SMS occurs in the controlled delayed MSNNs.

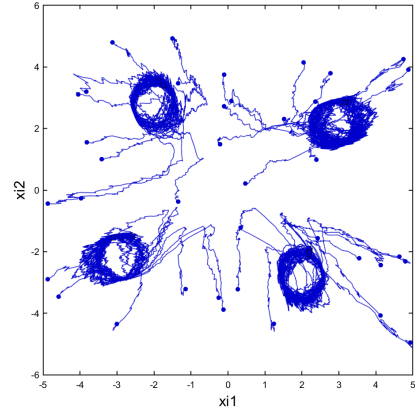


Fig. 1. Subsystem state evolution under periodic input.

IV. NUMERICAL EXAMPLES

A. Numerical Verification of Theoretical Results

To corroborate the theoretical analysis, a computational example is presented within this section.

Example 1: Consider a delayed MSNN system (1) featuring four interconnected subnetworks. Each subnetwork contains two neurons. The parameter matrices are defined as

$$\mathcal{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathcal{A} = \begin{bmatrix} 1 & 0.5 \\ -0.3 & 1 \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 1.5 & 0 \\ 0 & 1.5 \end{bmatrix}.$$

Then, consider the input periodic function as $I(t) = \begin{bmatrix} 0.8 \sin(t) \\ \cos(t) \end{bmatrix}$, the activation function $g(s) = \tanh(3s)$, time delay $\tau(t) = 2 - \sin(t)$, the sign vectors $\zeta = (\zeta_1, \dots, \zeta_n)^\top \in \{\pm 1\}^n$, and the diffusion coefficient of i th subnetwork is

$$\phi_i(x, y, E) = \begin{cases} \frac{0.25(\zeta_i x_i - 1)}{\zeta_i x_i} \left(-x_i + \sum_{j=1}^n \mathcal{B}_{ij} g_j(y_j) + I_i \right) & x \in \Delta_\zeta, y \in \mathbb{R}^2 \\ 0, x \notin \bigcup_{\zeta \in \{\pm 1\}^2} \Delta_\zeta, y \in \mathbb{R}^2 \end{cases}$$

which satisfy all the conditions in [36], so each subnetwork demonstrates mean-square exponential stability. Fig. 1 illustrates the time responses of two neurons in a single subnetwork when driven by a periodic input. The trajectories exhibit four distinct periodic orbits, confirming that each subnetwork possesses multiple locally exponentially stable periodic solutions, consistent with Proposition 1. This behavior verifies the intrinsic multistability of the constructed SNN model.

Under a constant input $I = [0.8, -0.8]^\top$, the neuron states converge to four stable equilibrium points, as shown in Fig. 2. These equilibria correspond to the static modes predicted by the theoretical analysis, validating that the proposed model admits 2^n mean-square stable equilibria for $n = 2$.

Now, establish the efficacy of our impulsive control strategy, which enables both dynamic and SMS in delayed MSNNs with fixed versus switching topological configurations.

Case I: Let us suppose

$$\mathbf{L} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

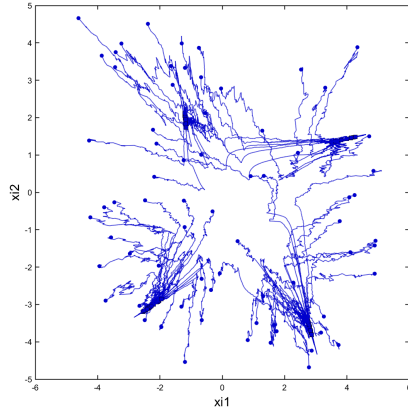
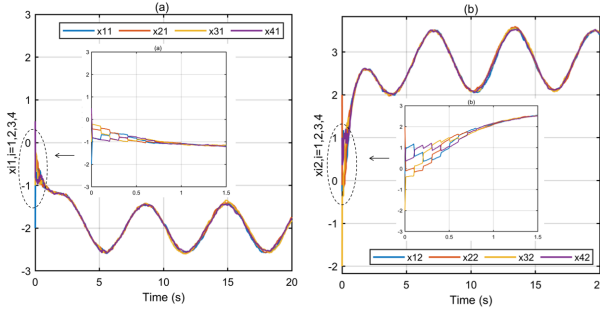


Fig. 2. Subsystem state evolution under constant input.

Fig. 3. Complete multisynchronization with fixed topology. (a) Synchronization behavior of the first state component $x_{1i}(t)$, $i = 1, 2, 3, 4$. (b) Synchronization behavior of the second state component $x_{2i}(t)$, $i = 1, 2, 3, 4$.

Employing an impulsive sequence defined by $T = 0.048$ and $T_a = 0.008$, it can be deduced that the parameters configured as follows: $N_0 = 1$, $k_0 = k_2 = 0.01$, $k_1 = 0.03$, $\rho = 1$, $\alpha = 0.91$, $\delta = 11$, $\gamma = 0.6$, $\Theta_1 = 0.08I_1$, $\Theta_2 = 0.07I_2$, $\varepsilon_0 = \varepsilon_2 = 1$, $\varepsilon_1 = 1$. Θ_1 and Θ_2 are chosen to satisfy the stability condition derived in Theorem 1, ensuring the feasibility of the LMI. Verification via MATLAB's LMI toolbox confirms that Theorem 1 (or Corollary 1) conditions are fulfilled, yielding the feasible solution

$$H = \begin{bmatrix} 0.5729 & -0.0012 \\ -0.0012 & 0.5754 \end{bmatrix} \quad Q_1 = \begin{bmatrix} 1.4742 & 0 \\ 0 & 1.5062 \end{bmatrix}$$

$$Q_2 = \begin{bmatrix} 0.2028 & 0 \\ 0 & 0.2038 \end{bmatrix}$$

and the coupling gain $\hat{d} = 0.58192$.

Simulation outcomes are depicted in Figs. 3–5 with initial states for these simulations uniformly randomized within $[-20, 20]$. Fig. 3 depicts the complete synchronization process of four coupled subnetworks for Example 1 under initial conditions $\xi = [-1, -0.5, 0.5, 1, -1, -2, 0.5, 1]^T$. All state trajectories converge to common manifolds, demonstrating the controller's capability to suppress stochastic perturbations and achieve complete synchronization.

Through the developed impulsive control scheme and the random selection of initial values, DMS and SMS are attained for the fixed-topology network (Figs. 4 and 5).

Fig. 4 shows that different subnetworks synchronize to four manifolds, verifying DMS. The result confirms Theorem 1

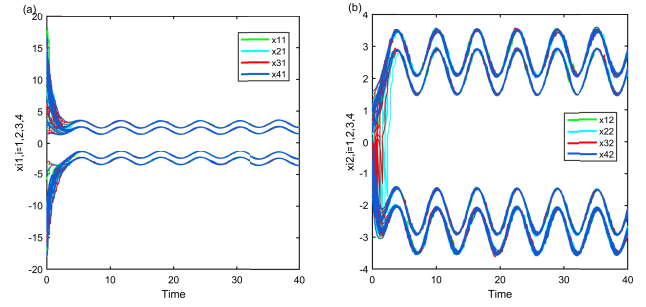


Fig. 4. DMS of MSNNs under fixed topology. (a) and (b) State trajectories of four subsystems under fixed topology.

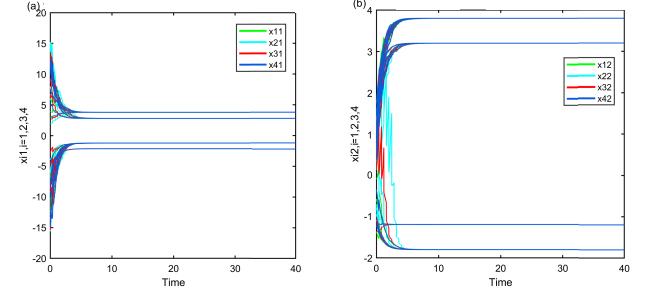


Fig. 5. SMS of MSNNs under fixed topology. (a) and (b) State trajectories of four subsystems under fixed topology.

that the impulsive sequence ensures exponential convergence of synchronization errors even in the presence of time-varying delays. Fig. 5 demonstrates SMS where subnetworks converge to different equilibrium points. This behavior validates Corollary 1 and highlights that the same control framework effectively handles both dynamic and static synchronization modes.

Case II: Four directed graphs are examined with corresponding Laplacian matrices

$$L_1 = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad L_2 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

$$L_3 = \begin{bmatrix} 2 & -1 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ -1 & 0 & 2 & -1 \\ -1 & -1 & 0 & 2 \end{bmatrix}$$

$$L_4 = \begin{bmatrix} 2 & -1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 2 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

The network topology undergoes stochastic switching every T among its four possible configurations. Verification via MATLAB's LMI toolbox confirms that Theorem 2 (or Corollary 2) conditions are satisfied, yielding the feasible solution

$$H = \begin{bmatrix} 0.5721 & -0.0012 \\ -0.0012 & 0.5746 \end{bmatrix}$$

and the coupling gains are

$$\hat{d}^1 = 0.67547, \quad \hat{d}^2 = 0.58115, \quad \hat{d}^3 = 0.39638, \quad \hat{d}^4 = 0.34047.$$

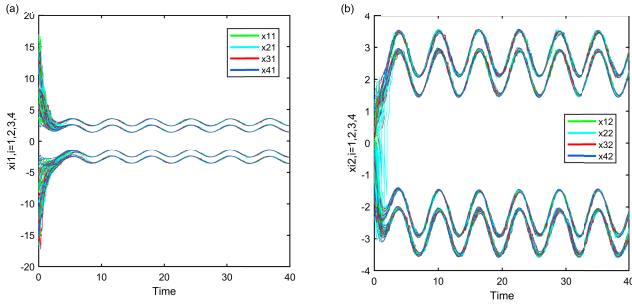


Fig. 6. DMS of MSNNs under switching topology. (a) and (b) State trajectories of four subsystems under switching topology.

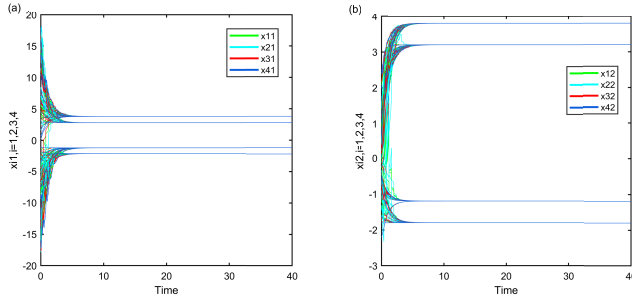


Fig. 7. SMS of MSNNs under switching topology. (a) and (b) State trajectories of four subsystems under switching topology.

To ensure generality, the initial values of variables are randomly selected in the $[-20, 20]$. In Fig. 6, the network topology switches randomly among four directed graphs. The trajectories remain synchronized within each manifold, confirming that the proposed control law maintains stability and multisynchronization even under rapidly varying communication structures, consistent with Theorem 2. Fig. 7 displays the convergence of all subnetworks to different stable equilibria under switching topologies. This verifies Corollary 2 and demonstrates the robustness of the coupled impulsive control scheme against topology variations.

B. Comparative Analysis

To illustrate how the proposed modeling framework more comprehensively captures the complex dynamic environment of real neural systems, we provide a comparative analysis with existing literature. Table I systematically contrasts the factors considered in the model of this article against those in several pioneering works on multisynchronization. The comparison focuses on the inclusion of four key elements: stochastic disturbances, time-varying delays, parameter uncertainties, and multisynchronization outcome.

The comparative analysis presented in Table I reveals the importance of the proposed framework's design, which is fundamentally architected to address the interconnected challenges of real-world systems. Unlike prior studies that often advance the field along isolated trajectories such as establishing multisynchronization in deterministic settings, incorporating stochasticity without delays, or developing robust control solely for mono-synchronization—this work integrates these facets into a unified model. This integrated

approach is critical, as practical systems from neuromorphic computing to cooperative robotics do not experience noise, delays, and uncertainties in isolation but are subject to their combined effects simultaneously. A controller designed for only one of these aspects is likely to fail when deployed in a genuine environment where all are present, thereby underscoring the necessity and importance of our holistic design.

Furthermore, to validate the superiority of the proposed control strategy in stochastic environments, comparative experiments were conducted using controller parameters from [31] and [32] under the same stochastic disturbance framework. It is important to emphasize that the controller parameters in [31] and [32] were designed for deterministic systems without stochastic disturbances, while the parameters in this article were optimized for systems with stochastic disturbances and parameter uncertainties. Using the method in [31], we can obtain

$$P = \begin{bmatrix} 1.7821 & -0.0967 \\ -0.0967 & 1.8197 \end{bmatrix}, \quad d = 0.2328.$$

Using the method in [32], we can obtain

$$P = \begin{bmatrix} 11.1231 & 2.4178 \\ 2.4178 & 14.7702 \end{bmatrix}, \quad S_1 = \begin{bmatrix} 1.3352 & 0 \\ 0 & 1.3352 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 1.3303 & 0 \\ 0 & 1.3303 \end{bmatrix}, \quad d = 0.2370.$$

Here, d represents control gains, like $\hat{d}$. P represents the Lyapunov function matrix, like H . The comparative results are shown in Fig. 8. It can be clearly observed that under various disturbance intensities, the synchronization error of our proposed method is significantly lower than both [31] and [32].

C. Parameter Sensitivity Analysis

To validate the robustness of the proposed control strategy under parameter uncertainties, we conducted systematic sensitivity analysis on the three types of parameter uncertainty bounds in system (1). The results are summarized in Table II.

The sensitivity analysis reveals distinct patterns in system robustness. Under mild uncertainty conditions ($k_0, k_2 \leq 0.01$), the system maintains excellent multisynchronization performance with final errors below 2×10^{-2} . The system exhibits the highest sensitivity to instantaneous connection matrix uncertainty (k_1), where even moderate uncertainty ($k_1 = 0.01$) causes significant performance degradation. In contrast, the system demonstrates remarkable robustness to delayed connection matrix uncertainty (k_2), maintaining excellent performance up to $k_2 = 0.05$. When any uncertainty bound exceeds 0.05, substantial performance deterioration occurs, characterized by final errors increasing by orders of magnitude. These findings provide practical guidance for control parameter tuning in uncertain environments, suggesting priority should be given to compensating for instantaneous connection uncertainties.

The derived sufficient condition (11) plays a pivotal role in guaranteeing multisynchronization of the considered MSNNs.

TABLE I
QUALITATIVE COMPARISON OF MODEL CAPABILITIES AND OUTCOMES

Model	Robustness Features		Disturbance & Synchronization	
	Time-varying Delays	Parameter Uncertainties	Stochastic Disturbances	Multisynchronization Outcome
[40]	Yes	No	No	Yes
[34]	No	No	No	Yes
[32]	Yes	Yes	No	Yes
[16]	No	No	Yes	No
[24]	Yes	No	Yes	No
(1)	Yes	Yes	Yes	Yes

Note: "Yes" indicates the feature is considered and handled effectively; "No" indicates the feature is not considered.

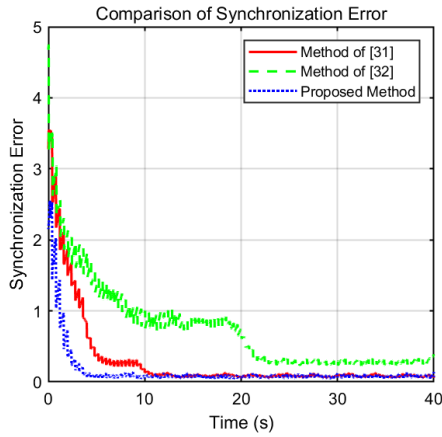


Fig. 8. Comparison of synchronization errors between the proposed method, [31], and [32] under random disturbances.

TABLE II
PARAMETER UNCERTAINTY SENSITIVITY ANALYSIS RESULTS

Uncertainty Type	Bound	Final Error	Convergence Rate
k_0 ($\Delta \mathcal{C}$)	0.0000	1.8096×10^{-2}	2.8102×10^{-1}
	0.0100	1.8540×10^{-2}	5.6184×10^{-2}
	0.0500	4.2828×10^{-1}	9.7096×10^{-2}
	0.1000	4.5491×10^0	3.6944×10^{-2}
k_1 ($\Delta \mathcal{A}$)	0.0000	1.9858×10^{-2}	1.9003×10^{-1}
	0.0100	4.8083×10^{-1}	1.0239×10^{-1}
	0.0500	3.6367×10^0	2.7909×10^{-2}
	0.1000	7.3270×10^0	2.8433×10^{-2}
k_2 ($\Delta \mathcal{B}$)	0.0000	1.5833×10^{-2}	1.9605×10^{-1}
	0.0100	1.7182×10^{-2}	2.4550×10^{-1}
	0.0500	1.8507×10^{-2}	2.7923×10^{-1}
	0.1000	4.2315×10^0	1.7010×10^{-2}

To provide deeper insights into the control design, we conduct a comprehensive analysis to elucidate the influence of key parameters on the stability condition.

Fig. 9 illustrates the variation of the sufficient stability condition with respect to the key design parameters. In Fig. 9(a), decreasing α enhances the impulsive contraction effect, leading to a larger stability margin. In Fig. 9(b), the parameter δ , representing the state decay rate, exhibits high negative sensitivity. In Fig. 9(c), larger γ improves robustness against time delay but gradually drives the stability condition toward zero, reflecting the conservative tradeoff between robustness and feasibility. These curves clearly demonstrate how the three parameters interact mathematically in the sufficient condition

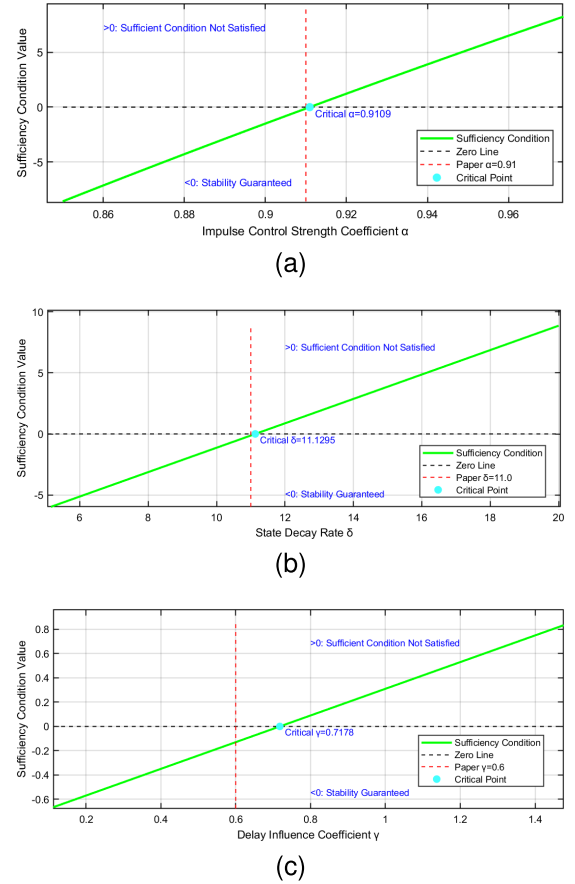


Fig. 9. Effects of different parameters on the sufficient stability condition. (a) Effect of α on the sufficient stability condition. (b) Effect of δ on the sufficient stability condition. (c) Effect of γ on the sufficient stability condition.

(11), providing intuitive guidelines for parameter tuning. For practical tuning, we recommend initializing $\alpha \in [0.8, 0.95]$, δ , and γ with moderate positive values, then iteratively adjusting based on sensitivity tests such as increasing δ or decreasing α to accelerate convergence, but reducing δ or γ if oscillations occur. This systematic analysis bridges theoretical parameter roles with practical design guidelines, enhancing the applicability and interpretability of the proposed control framework.

Remark 6: The proposed framework enhances generality and adaptability through key theoretical advances. By removing the restrictive assumption of delay differentiability, it supports systems with sharp or irregular timing variations,

which is essential in digital communications with packet loss or event-triggered control. Its inherent support for switching topologies further enables adaptability in dynamic networks like mobile ad hoc or multiagent systems.

These features collectively bridge theoretical models and practical needs, reinforcing the application-oriented nature of this work. The introduced multisynchronization capability under stochasticity is vital in areas such as heterogeneous robotics, neuromorphic computing, and secure communications, where multiple subsystems must coordinate distinct tasks amid uncertainty. As summarized in Table I, the framework meets demands of advanced domains: in neuromorphic computing, it allows processor clusters to synchronize under fluctuating workloads; in multirobot systems, it enables subteams to maintain coordination despite sensor noise, delays, and changing network conditions.

By capturing a broad spectrum of real-world complexities, this framework moves beyond incremental theory. It establishes a versatile and robust foundation for controlling complex systems in stochastic, uncertain, and dynamic engineering environments.

V. CONCLUSION

This article has presented a comprehensive study on multisynchronization control for a class of MSNNs subject to time-varying delays, uncertainties, and stochastic disturbances. The main contribution lies in the design of a coupled impulsive controller that effectively achieves multisynchronization under stochastic perturbation. Leveraging Lyapunov analysis and LMI techniques, we have established easy-to-verify synchronization criteria for both fixed and switching topologies.

Despite these advancements, our approach has limitations, including potential conservatism in the stability conditions and the need for continuous state information. Future work will focus on extending the control scheme to more complex network structures such as memristive or multilayer networks, designing distributed adaptive impulsive protocols, and enhancing robustness to topology uncertainties and switching errors. These extensions are expected to further broaden the applicability between theoretical impulsive control and real-world neural systems.

REFERENCES

- [1] H.-L. Wei, S. A. Billings, Y. Zhao, and L. Guo, "Lattice dynamical wavelet neural networks implemented using particle swarm optimization for spatio-temporal system identification," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 181–185, Jan. 2009.
- [2] P. Z. Marmarelis and K.-I. Naka, "Identification of multi-input biological systems," *IEEE Trans. Biomed. Eng.*, vols. BME-21, no. 2, pp. 88–101, Mar. 1974.
- [3] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, no. 6684, pp. 440–442, Jun. 1998.
- [4] P. Cheridito, A. Jentzen, and F. Rossmannek, "Efficient approximation of high-dimensional functions with neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 7, pp. 3079–3093, Jul. 2022.
- [5] Y. Liu, X. Chen, Y. Wu, H. Cai, and H. Yokoi, "Adaptive neural network control of a flexible spacecraft subject to input nonlinearity and asymmetric output constraint," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6226–6234, Nov. 2022.
- [6] H. Yu, S. Zhu, S. Wen, and C. Mu, "Finite-time stability in probability of stochastic delay systems via generalized Halanay inequality," *Syst. Control Lett.*, vol. 195, Jan. 2025, Art. no. 105969.
- [7] R. Sengupta, I. Polian, and J. P. Hayes, "WASENN: Wavelet assisted stochastic enabled neural network for human activity recognition," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 10, pp. 5761–5773, Oct. 2025.
- [8] H. Shen, C. Peng, H. Yan, and S. Xu, "Data-driven near optimization for fast sampling singularly perturbed systems," *IEEE Trans. Autom. Control*, vol. 69, no. 7, pp. 4689–4694, Jul. 2024.
- [9] H. Shen, Y. Wang, H. Yan, and S. Xu, "Data-driven single-loop policy iteration control of uncertain singularly perturbed systems," *IEEE Trans. Autom. Control*, vol. 70, no. 12, pp. 8314–8320, Dec. 2025.
- [10] Y. Chen, S. Zhu, H. Yan, M. Shen, X. Liu, and S. Wen, "Event-based global exponential synchronization for quaternion-valued fuzzy memristor neural networks with time-varying delays," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 3, pp. 989–999, Mar. 2024.
- [11] Q. Zhao, S. Zhu, Z. Zhang, W. Luo, and S. Wen, "Multistability of almost periodic solutions for fuzzy competitive NNs with time-varying delays," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 11835–11846, Jul. 2025.
- [12] X. Zhang, C. Li, H. Li, and Z. Cao, "Synchronization of uncertain coupled neural networks with time-varying delay of unknown bound via distributed delayed impulsive control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3624–3635, Jul. 2023.
- [13] Y. Liu, Z. Wang, X. Huang, and H. Shen, "Finite-time multisynchronization of coupled discontinuous cohen-grossberg neural networks and its application in associative memory," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 11, pp. 7109–7119, Nov. 2025.
- [14] S. Zhu, H. Sang, K. Zhang, F. Kong, and J. Lü, "Synchronization of intermittently coupled neural networks with coupling delay," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 7984–7996, May 2025.
- [15] H. Long, J. Ci, Z. Guo, S. Wen, and T. Huang, "Synchronization of coupled switched neural networks subject to hybrid stochastic disturbances," *Neural Netw.*, vol. 166, pp. 459–470, Sep. 2023.
- [16] Z. Cao, C. Li, Z. He, X. Zhang, and L. You, "Synchronization of coupled stochastic reaction-diffusion neural networks with multiple weights and delays via pinning impulsive control," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 2, pp. 820–833, Mar. 2022.
- [17] X. Ji, J. Lu, B. Jiang, and K. Shi, "Distributed synchronization of delayed neural networks: Delay-dependent hybrid impulsive control," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 2, pp. 634–647, Mar. 2022.
- [18] Y. Chen, S. Zhu, Y. Li, X. Liu, and S. Wen, "Finite-time synchronization for delayed NNs via generalized Halanay inequalities," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 55, no. 7, pp. 4742–4751, Jul. 2025.
- [19] Q. Qiu and H. Su, "Finite-time output synchronization for output-coupled reaction-diffusion neural networks with directed topology," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 3, pp. 1386–1394, May 2022.
- [20] W. Yao, C. Wang, Y. Sun, and C. Zhou, "Robust multimode function synchronization of memristive neural networks with parameter perturbations and time-varying delays," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 1, pp. 260–274, Jan. 2022.
- [21] F. Li, Y.-N. Wang, and H. Shen, "Dual event-triggered synchronization of two-time-scale jumping neural networks and its application in image encryption and decryption," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 2, pp. 943–954, Mar. 2025.
- [22] W. Sun, Z. Tang, J. Feng, and J. H. Park, "Event-triggered impulsive synchronization for neural networks subject to deception attacks: Saturation control and optimization," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 12071–12081, 2025.
- [23] C. Jiang, Z. Tang, J. H. Park, and J. Feng, "Matrix measure-based event-triggered impulsive quasi-synchronization on coupled neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1821–1832, Feb. 2024.
- [24] Z. Tang, C. Jiang, Y. Wang, J. Feng, and J. H. Park, "Average impulsive weight based event-triggered impulsive synchronization on coupled neural networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 4, pp. 2180–2189, Jul. 2023.
- [25] K. Gao, J. Lu, W. X. Zheng, and X. Chen, "Synchronization in coupled neural networks with hybrid delayed impulses: Average impulsive delay-gain method," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3608–3617, Feb. 2025.
- [26] J. Ping, S. Zhu, W. Luo, and Z. Zhang, "Hyper-exponential stabilization of neural networks by event-triggered impulsive control with actuation delay," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7778–7781, Apr. 2025.

- [27] D. Ding, Z. Tang, C. Wen, Z. Ji, and J. H. Park, "Secure impulsive synchronization for complex networks under fully distributed deception attacks: Error conjunction approach," *IEEE Trans. Ind. Informat.*, vol. 21, no. 12, pp. 9435–9445, Dec. 2025.
- [28] L. Peng, X. Li, D. Bi, X. Xie, and Y. Xie, "Multiple μ -stable synchronization control for coupled memristive neural networks with unbounded time delays," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 2, pp. 990–1002, Feb. 2022.
- [29] W.-H. Chen, X. Lu, and W. X. Zheng, "Impulsive stabilization and impulsive synchronization of discrete-time delayed neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 4, pp. 734–748, Apr. 2015.
- [30] L. Zhang, K. Shi, J. Lu, and J. Lou, "Bipartite synchronization of antagonistic coupled neural networks: Average-delay pinning impulsive control," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 9, pp. 3814–3818, Sep. 2022.
- [31] Y.-W. Wang, W. Yang, J.-W. Xiao, and Z.-G. Zeng, "Impulsive multisynchronization of coupled multistable neural networks with time-varying delay," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 7, pp. 1560–1571, Jul. 2017.
- [32] X. Lv, X. Li, J. Cao, and M. Perc, "Dynamical and static multisynchronization of coupled multistable neural networks via impulsive control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 6062–6072, Dec. 2018.
- [33] X. Lv, J. Cao, X. Li, M. Abdel-Aty, and U. A. Al-Juboori, "Synchronization analysis for complex dynamical networks with coupling delay via event-triggered delayed impulsive control," *IEEE Trans. Cybern.*, vol. 51, no. 11, pp. 5269–5278, Nov. 2021.
- [34] Y. Liu, Z. Wang, X. Huang, and H. Shen, "Multisynchronization of coupled multistable neural networks via event-triggered impulsive control and its application to associative memory," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 6729–6739, 2025.
- [35] C. Godsil and G. Royle, *Algebraic Graph Theory*. New York, NY, USA: Springer, 2013.
- [36] W.-H. Chen, S. Luo, and X. Lu, "Multistability in a class of stochastic delayed Hopfield neural networks," *Neural Netw.*, vol. 68, pp. 52–61, Aug. 2015.
- [37] J. Lu, D. W. C. Ho, and J. Cao, "A unified synchronization criterion for impulsive dynamical networks," *Automatica*, vol. 46, no. 7, pp. 1215–1221, Jul. 2010.
- [38] S. Boyd, L. E. Ghaoui, E. Feron, and V. Balakrishnan, "Linear matrix," in *Inequalities in System and Control Theory*. Philadelphia, PA, USA: SIAM, 1994.
- [39] Z.-H. Guan, Z.-W. Liu, G. Feng, and Y.-W. Wang, "Synchronization of complex dynamical networks with time-varying delays via impulsive distributed control," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 8, pp. 2182–2195, Aug. 2010.
- [40] Z. Wang, Y. Liu, X. Huang, and H. Shen, "Quasi-multisynchronization and quasi-monosynchronization of delayed neural networks with parameter mismatch via impulsive control," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 14646–14656, 2025.



interests include neural networks and stochastic control.

Song Zhu was born in 1982. He received the B.S. degree in mathematics from Jiangsu Normal University, Xuzhou, China, in 2004, and the M.S. degree in probability and mathematical statistics and the Ph.D. degree in system engineering from Huazhong University of Science and Technology, Wuhan, China, in 2007 and 2010, respectively.

He is currently a Professor with the School of Mathematics, China University of Mining and Technology, Xuzhou. He has published over 90 international journal papers. His current research

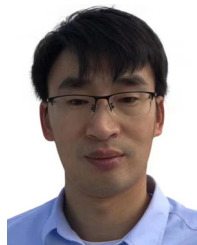


Zhen Zhang received the Ph.D. degree from China University of Mining and Technology, Xuzhou, China, in 2022. He is currently a Post-Doctoral Fellow with China University of Mining and Technology. His research interests include nonlinear control theory and its applications in servo systems.



Weiwei Luo received the Ph.D. degree from the Department of Control Science and Engineering, Harbin Institute of Technology, Harbin, China, in 2020.

He is currently a Lecturer with the School of Mathematics, China University of Mining and Technology, Xuzhou, China. His research interests include time-varying systems, spacecraft control, and bounded and nonlinear control.



Shiping Wen (Senior Member, IEEE) received the M.Eng. degree in control science and engineering from the School of Automation, Wuhan University of Technology, Wuhan, China, in 2010, and the Ph.D. degree in control science and engineering from the School of Automation, Huazhong University of Science and Technology, Wuhan, in 2013.

He is currently a Professor with the Australian Artificial Intelligence Institute (AAIL), University of Technology Sydney, Sydney, NSW, Australia. His research interests include neural networks, deep

learning, computer vision, and their applications in medical informatics.

Dr. Wen is a fellow of British Computer Society (BCS). He received the 2017 Young Investigator Award from the Asian Pacific Neural Network Association and the 2015 Chinese Association of Artificial Intelligence Outstanding Ph.D. Dissertation Award. He was listed as a Highly Cited Researcher by Clarivate Analytics in 2018 and 2020, respectively. He currently serves as an Associate Editor for *Knowledge-Based Systems*, *Engineering Applications of AI*, and *Neural Processing Letters*, and the Leading Guest Editor for *IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING* and *Sustainable Cities and Society*.



Xuemei Li was born in 2002. She received the B.S. degree in mathematics from Nantong University, Nantong, China, in 2024. She is currently pursuing the M.S. degree in mathematics with the School of Mathematics, China University of Mining and Technology, Xuzhou, China.

Her research interests include the dynamics of neural networks and control systems.